

Computer = Data Processor

Data Processor → Data Processing

① What is Operating System?

→ A set of programs which act as an interface between user interface and computer hardware & controls the execution of all programs.

② Functions of Operating Systems -

- Memory Management
- Processor Management
- Device Management
- File Management
- Security
- Control over system performance
- Job accounting
- Error detecting aids
- Coordination between other softwares & users

③ Working of Operating System

(i) User View

- ease of use & resource utilization
- from user view, OS is designed to maximize resource utilization, and to assure all available CPU time, memory and I/O devices are used efficiently.

(ii) System View

- from system's view, OS is the program ~~is~~ involved with hardware
- resource allocation
- manages CPU time, memory space, file storage space & I/O devices.
- act as a control program which manages the execution of the user program to prevent errors & improper use of the computer.

④ Advantages of Operating System

- (i) Convenience - easy to use commands, graphical interfaces
- (ii) Efficiency - utilizes computer resources

⑤ Generations of Operating Systems

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① First Generation (1940 - early 1950's)

- vacuum tubes
- no protection
- machine language
- no real operating system
- heated up systems

② Second Generation (1955 - 1965)

- GIMOSIS : first operating system
- IBM General motors
- transistors
- batch operating system, punch cards
- minimum protection
- base for multi-programming

③ Third Generation (1965 - 1980)

- Integrated Circuits (IC)
- Multi-programming and multi-tasking

④ Fourth Generation (1980 - present)

- Windows, MacOS
- Provided GUI
- computer networking
- maximum CPU utilization

⑥ Types of Operating Systems

1. Batch OS

- Batch OS is designed to manage and execute a large number of jobs with similar needs by processing them into groups.
- Advantages : Multiple users can share the batch systems.
Idle time is less
- Disadvantages : Hard to debug
Sometimes costly
Difficult to accurately predict the time required for a job to complete while it is in the queue.

2. Multi-programming OS

- Multi-programming OS can be simply illustrated as more than one program is present in the main memory and any one of them can be kept in execution

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- Advantages: Increases the Throughput of the system
helps in reducing the response time
- Disadvantages: No facility for user interaction of system resources with the system.

3. Multi-tasking Programming OS

- Multi-tasking programming is simply an extension of multi-programming OS with having Round-Robin Scheduling Algorithm.
- It can run multiple programs simultaneously. Allows the user to run more than one program at a time.
- Two types: Pre-emptive & Co-operative
- Advantages: Multiple programs can be executed
comes with proper memory management
- Disadvantages: System gets heated in case of heavy programs

4. Time-Sharing OS

- Multi-tasking
- Each task gets a particular time limit (Quantum) to get executed, then the system moves to the next task.
- Advantages: equal time opportunity
CPU idle time reduced
- Disadvantages: Reliability problem
Complexity and security risks.

5. Real-time OS

- In real-time OS, each job carries a certain deadline within which the job is supposed to be completed or there will be huge loss.
- Military Applications: Missiles (Hard Real-Time OS)
- Soft Real-Time OS: where time constraint is less strict.
- Advantages: Maximum utilization of devices and systems
- Disadvantages: Costly to develop
very complex & critical

6. Distributed OS

- Distributed OS is not installed on a single machine, it's divided into parts & these parts are loaded into different devices.
- Advantages: Failure of one will not affect the others
Easily scalable
- Disadvantages: Failure of main network will stop entire communication
Not well-defined language yet developed

- Disadvantages: Failure of main network will stop entire communication
Not well-defined language yet developed
Not readily available as they're very expensive
- Many networks, global

7. Clustered OS

- One network, local

8. Embedded OS

- Fixed functionalities
- Washing Machine

⑦ Structure of Operating Systems

1. Simple Structure

- Simple OS doesn't have well-defined structure if are small, simple and limited systems.
- MS-DOS
- Advantages - Few interfaces between Application & Hardware
- Easy for kernel developers
- Disadvantages - Doesn't enforce data hiding
No clear boundaries btw modules

2. Monolithic Structure

- The OS kernel is designed to provide all OS services like process management, memory management, in a single, large binary.
- All code runs in kernel space, no separation btw kernel and user-level interface
- UNIX-OS
- Advantages: High performance coz of direct kernel contact
- Disadvantages: Less secure & less stable OS.

3. Layered Structure

- OS is broken into number of layers (levels); bottom(0) is hardware and top-most is user-interface.
(layer N) (layer 0)
- Advantages: Easy debugging
Easier to enhance OS on a single level without disturbing the other levels
- Disadvantages: Application performance degraded as compared to simple structure.
Requires careful planning for designing of layers

Simple structure.
Requires careful planning for designing of layers

4. Microkernel Structure

- Removing all non-essential components from the kernel and implementing them as system and user programs. This results in a smaller kernel - called Microkernel.
- Advantages : All new services can be added to the user space and doesn't require kernel to be modified.
More secure and reliable
- Disadvantages : Increased level of inter-module communication degrades system performance.

5. Modular Structure

- Considered best approach for an OS.
- Involves designing of modular kernel.
- resembles layered structure as each kernel has defined & protected interfaces but more flexible than layered structure because each module can call any other module.

6. Hybrid-kernel Structure

- Nothing but a combination of Monolithic and Micro-kernel.
- Implements speed of design of monolithic and modularity and stability of Microkernel.
- Advantages : Advantages of both structures
Supports wide range of hardware & softwares
- Disadvantages : Increases overall complexity
Increases time complexity & decreases performance compared to Monolithic

⑧ OS Services

- | | |
|----------------------|---------------------------|
| - Memory Management | - File Management |
| - User Interface | - Program Execution |
| - Security & Privacy | - Process Management |
| - Networking | - Resource Management |
| - Error Handling | - Input Output Operations |